

MA4261

Differential Geometry of Curves and Surfaces

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1 Introduction to Curves

2 Introduction to Surfaces

3 Gauss' Curvature and Second Fundamental Form

3.1 Curvature of Surfaces

In Chapter 1, we defined curvature as $\kappa = |\alpha''(s)|$ — curvature **at a point**. In this chapter, we try to find a definition for curvature of a **surface at a point**.

Gauss had this idea: given a point $p \in S$ on a regular surface S , along the normal vector $\mathbf{N}$ perpendicular to the surface at $\mathbf{p}$, we take a knife and cut the surface downward along the tangent direction $\mathbf{v}$. The curvature of the resulting cross-sectional curve gives us information about the curvature of the surface at $\mathbf{p}$.

3.2 Linear Algebra Review

First, we define a *self-adjoint* linear transformation.

Definition 3.1 (Self-adjoint map). Let V be an inner product space, and $T : V \rightarrow V$ a linear transformation. Then T is **self-adjoint** if

$$\langle T(\mathbf{v}), \mathbf{w} \rangle = \langle \mathbf{v}, T(\mathbf{w}) \rangle \quad \text{for all } \mathbf{v}, \mathbf{w} \in V.$$

Lemma 3.2 (Self-adjoint via basis). Suppose we have a basis $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ of V . Then it suffices to check

$$\langle T(\mathbf{x}_1), \mathbf{x}_2 \rangle = \langle \mathbf{x}_1, T(\mathbf{x}_2) \rangle.$$

The usefulness of this lemma is that if we *do* have a basis, we only need to check self-adjointness on the basis vectors rather than all $\mathbf{v}, \mathbf{w} \in V$.

Definition 3.3 (Matrix representation of a linear transformation). Let $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ be a basis of V , and $T : V \rightarrow V$ a linear transformation. Suppose

$$T(\mathbf{x}_1) = a\mathbf{x}_1 + b\mathbf{x}_2, \quad T(\mathbf{x}_2) = c\mathbf{x}_1 + d\mathbf{x}_2.$$

Then the matrix representation of T with respect to $\mathcal{B}$ is

$$[T]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Lemma 3.4 (Self-adjointness via matrix representation). Let $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ be an orthonormal basis of V . Then $T : V \rightarrow V$ is self-adjoint if and only if

$$[T]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is symmetric, i.e. $b = c$.

Proof. Let $[T]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\langle T(\mathbf{x}_1), \mathbf{x}_2 \rangle = \langle a\mathbf{x}_1 + b\mathbf{x}_2, \mathbf{x}_2 \rangle = b, \quad \langle \mathbf{x}_1, T(\mathbf{x}_2) \rangle = \langle \mathbf{x}_1, c\mathbf{x}_1 + d\mathbf{x}_2 \rangle = c.$$

Hence $\langle T(\mathbf{x}_1), \mathbf{x}_2 \rangle = \langle \mathbf{x}_1, T(\mathbf{x}_2) \rangle \iff b = c$, which proves both directions. $\square$

Lemma 3.5 (Existence of eigenvalue pairs). *Let $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ be a real 2×2 symmetric matrix. The characteristic polynomial of A is*

$$p(\lambda) = \det(\lambda I - A) = \lambda^2 - (a + d)\lambda + (ad - b^2),$$

with real eigenvalues

$$\lambda_1, \lambda_2 = \frac{a + d}{2} \pm \sqrt{\left(\frac{a - d}{2}\right)^2 + b^2} \in \mathbb{R}.$$

The corresponding eigenvectors are found via

$$E_{\lambda_i} = \text{Null}(\lambda_i I_2 - A) = \{\mathbf{v} \in \mathbb{R}^2 \mid (\lambda_i I_2 - A)\mathbf{v} = \mathbf{0}\}.$$

Let $\mathbf{v}_1$ be the unit eigenvector for λ_1 , and let $\mathbf{v}_2$ be the unit vector perpendicular to $\mathbf{v}_1$. Then $\mathbf{v}_2$ is an eigenvector of A with eigenvalue λ_2 .

Proof. Since $A = A^\top$,

$$(A\mathbf{v}_2) \cdot \mathbf{v}_1 = \mathbf{v}_2^\top A^\top \mathbf{v}_1 = \mathbf{v}_2^\top (A\mathbf{v}_1) = \lambda_1 \mathbf{v}_2^\top \mathbf{v}_1 = 0.$$

Hence $A\mathbf{v}_2 \perp \mathbf{v}_1$, so $A\mathbf{v}_2$ is parallel to $\mathbf{v}_2$, i.e. $A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$. □

Corollary 3.6 (Self-adjoint maps admit an orthonormal eigenbasis). *Given a real symmetric 2×2 matrix, there exists an orthonormal basis of real eigenvectors*

$$\left\{ \begin{pmatrix} p_{11} \\ p_{21} \end{pmatrix}, \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} \right\}$$

such that $P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}$ is an orthogonal matrix, i.e. $P^\top = P^{-1}$.

Theorem 3.7 (Main theorem on self-adjoint matrices). *Given a real symmetric matrix A , we can diagonalise*

$$A = PDP^{-1} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}^{-1}.$$

3.3 The Gauss Map

Consider a regular surface S and a point $\mathbf{q} \in S$. We can choose a unit vector $N(\mathbf{q})$ perpendicular to the tangent plane $T_{\mathbf{q}}(S)$:

$$N(\mathbf{q}) = \frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}, \quad \text{or} \quad N(\mathbf{q}) = -\frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}.$$

Since $N(\mathbf{q}) \in S^2$ for each $\mathbf{q}$, the Gauss map sends $\mathbf{q} \mapsto N(\mathbf{q})$.

Definition 3.8 (Gauss Map). Let S be a regular surface. The **Gauss map** is a smooth map

$$N : S \rightarrow S^2$$

where $N(\mathbf{q})$ is the unit normal at $\mathbf{q}$ as above.

Remarks.

1. Not every regular surface admits a Gauss map.

2. $N(\mathbf{q})$ can always *locally* be found via the cross product of coordinate vectors, which exists by regularity of S .
3. If $N : S \rightarrow S^2$ is a Gauss map, then $-N : S \rightarrow S^2$ is also a Gauss map, corresponding to the other consistent choice of unit normal.

3.4 The Second Fundamental Form

Consider the map $dN : T_p(S) \rightarrow T_{N(p)}(S^2)$. Since $\mathbf{q}$ is the origin of $T_{\mathbf{q}}(S)$ and $N(\mathbf{q})$ is the origin of $T_{N(\mathbf{q})}(S^2)$, both tangent planes contain the origin and are perpendicular to $N(\mathbf{q})$. Hence the two tangent spaces coincide, and we may view dN as a linear transformation

$$dN : T_p(S) \rightarrow T_p(S).$$

Proposition 3.9. *dN is a self-adjoint linear transformation.*

Proof. By the basis lemma, it suffices to show $\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = \langle \mathbf{x}_u, dN(\mathbf{x}_v) \rangle$. Since $N \circ \mathbf{x} \perp \mathbf{x}_u$, we have $\langle N \circ \mathbf{x}, \mathbf{x}_u \rangle = 0$. Differentiating with respect to v :

$$\left\langle \frac{\partial}{\partial v}(N \circ \mathbf{x}), \mathbf{x}_u \right\rangle + \langle N \circ \mathbf{x}, \mathbf{x}_{uv} \rangle = 0.$$

Since $\frac{\partial}{\partial v}(N \circ \mathbf{x}) = dN(\mathbf{x}_v)$, this gives $\langle dN(\mathbf{x}_v), \mathbf{x}_u \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{uv} \rangle$. Interchanging u and v : $\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{vu} \rangle$. Since $\mathbf{x}_{uv} = \mathbf{x}_{vu}$, we conclude

$$\langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle = -\langle N \circ \mathbf{x}, \mathbf{x}_{uv} \rangle = \langle \mathbf{x}_u, dN(\mathbf{x}_v) \rangle. \quad \square$$

Definition 3.10 (Second Fundamental Form). For $\mathbf{v} \in T_p(S)$, the **second fundamental form** of S at $\mathbf{q}$ is

$$\text{II}_{\mathbf{q}}(\mathbf{v}) = -\langle dN(\mathbf{v}), \mathbf{v} \rangle.$$

Proposition 3.11 (Finding dN from II_p). Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be an orthonormal basis of $T_p(S)$, and write

$$-[dN]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$a = \text{II}_{\mathbf{q}}(\mathbf{v}_1), \quad d = \text{II}_{\mathbf{q}}(\mathbf{v}_2), \quad b = \frac{1}{2} [\text{II}_{\mathbf{q}}(\mathbf{v}_1 + \mathbf{v}_2) - \text{II}_{\mathbf{q}}(\mathbf{v}_1) - \text{II}_{\mathbf{q}}(\mathbf{v}_2)].$$

Proof. From the matrix representation, $-dN(\mathbf{v}_1) = a\mathbf{v}_1 + b\mathbf{v}_2$ and $-dN(\mathbf{v}_2) = b\mathbf{v}_1 + d\mathbf{v}_2$. Then by definition of II :

$$\text{II}_{\mathbf{q}}(\mathbf{v}_1) = \langle a\mathbf{v}_1 + b\mathbf{v}_2, \mathbf{v}_1 \rangle = a, \quad \text{II}_{\mathbf{q}}(\mathbf{v}_2) = \langle b\mathbf{v}_1 + d\mathbf{v}_2, \mathbf{v}_2 \rangle = d.$$

For b , expand:

$$\begin{aligned} \text{II}_{\mathbf{q}}(\mathbf{v}_1 + \mathbf{v}_2) &= \langle -dN(\mathbf{v}_1 + \mathbf{v}_2), \mathbf{v}_1 + \mathbf{v}_2 \rangle \\ &= \langle -dN(\mathbf{v}_1), \mathbf{v}_1 \rangle + \langle -dN(\mathbf{v}_1), \mathbf{v}_2 \rangle + \langle -dN(\mathbf{v}_2), \mathbf{v}_1 \rangle + \langle -dN(\mathbf{v}_2), \mathbf{v}_2 \rangle \\ &= \text{II}_{\mathbf{q}}(\mathbf{v}_1) + \text{II}_{\mathbf{q}}(\mathbf{v}_2) + 2b. \end{aligned}$$

Solving for b gives the stated formula. $\square$

3.5 Normal Curvature

Definition 3.12 (Normal Curvature). Let $\alpha(s)$ be a curve parameterised by arc length on a surface S , passing through $\mathbf{p} = \alpha(0)$. The **normal curvature** is

$$k_n = \kappa \cos \theta,$$

where θ is the angle between $\alpha''(0)$ and $N(\mathbf{p})$.

The normal curvature is the projection of the curvature vector $\alpha''(0)$ onto the surface normal. In particular, $|k_n(\mathbf{p})| \leq \kappa$.

Proposition 3.13 (Normal curvature = second fundamental form). Let $\alpha(s)$ be a curve parameterised by arc length, and $\mathbf{p} = \alpha(0) \in S$. Then

$$\Pi_{\mathbf{p}}(\alpha'(0)) = -\langle dN(\alpha'(0)), \alpha'(0) \rangle = k_n(\mathbf{p}).$$

Proof. Since $N(\alpha(s)) \perp \alpha'(s)$, we have $\langle N(\alpha(s)), \alpha'(s) \rangle = 0$. Differentiating:

$$\begin{aligned} 0 &= \frac{d}{ds} \langle N(\alpha(s)), \alpha'(s) \rangle \\ &= \langle dN(\alpha'(s)), \alpha'(s) \rangle + \langle N(\alpha(s)), \alpha''(s) \rangle \\ &= -\Pi_{\mathbf{p}}(\alpha'(s)) + \langle N(\alpha(s)), \kappa \mathbf{n}(s) \rangle \\ &= -\Pi_{\mathbf{p}}(\alpha'(s)) + \kappa \cos \theta. \end{aligned}$$

Hence $\Pi_{\mathbf{p}}(\alpha'(s)) = \kappa \cos \theta = k_n(\mathbf{p})$. □

The significance is that the normal curvature depends only on the unit tangent vector $\alpha'(0)$: if two curves satisfy $\alpha'(0) = \beta'(0)$, they have the same normal curvature.

3.6 Gaussian Curvature

Proposition 3.14. $-dN : T_{\mathbf{p}}(S) \rightarrow T_{\mathbf{p}}(S)$ is a self-adjoint linear transformation.

Proof. Since dN is self-adjoint, for all $\mathbf{v}, \mathbf{w} \in T_{\mathbf{p}}(S)$:

$$\langle -dN(\mathbf{v}), \mathbf{w} \rangle = -\langle dN(\mathbf{v}), \mathbf{w} \rangle = -\langle \mathbf{v}, dN(\mathbf{w}) \rangle = \langle \mathbf{v}, -dN(\mathbf{w}) \rangle. \quad \square$$

Definition 3.15 (Principal curvatures and directions). By the main theorem on self-adjoint maps, there exists an orthonormal basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ of $T_{\mathbf{p}}(S)$ diagonalising $-dN$:

$$-[dN]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}.$$

We call $\mathbf{v}_1, \mathbf{v}_2$ the **principal directions** and k_1, k_2 the **principal curvatures** at $\mathbf{p}$.

Proposition 3.16 (Second fundamental form in principal directions). For $\mathbf{v} = a\mathbf{v}_1 + b\mathbf{v}_2 \in T_{\mathbf{p}}(S)$,

$$\Pi_{\mathbf{p}}(\mathbf{v}) = k_1 a^2 + k_2 b^2.$$

Proof.

$$\begin{aligned} \Pi_{\mathbf{p}}(\mathbf{v}) &= \langle -dN(a\mathbf{v}_1 + b\mathbf{v}_2), a\mathbf{v}_1 + b\mathbf{v}_2 \rangle \\ &= \langle k_1 a\mathbf{v}_1 + k_2 b\mathbf{v}_2, a\mathbf{v}_1 + b\mathbf{v}_2 \rangle \\ &= k_1 a^2 + k_2 b^2. \end{aligned}$$

□

Proposition 3.17. The maximal and minimal normal curvatures at $\mathbf{p}$ occur at $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively, and they are perpendicular to each other.

Definition 3.18 (Gaussian and mean curvatures). Let k_1, k_2 be the principal curvatures at $\mathbf{p}$. The **Gaussian curvature** and **mean curvature** are defined as

$$K(\mathbf{p}) = \det(-[dN_p]_{\mathcal{B},\mathcal{B}}) = k_1 k_2, \quad H(\mathbf{p}) = \frac{1}{2} \operatorname{tr}(-[dN_p]_{\mathcal{B},\mathcal{B}}) = \frac{k_1 + k_2}{2}.$$

Remarks.

1. $K(\mathbf{p})$ is the determinant of $-dN$ (product of eigenvalues); $2H(\mathbf{p})$ is the trace (sum of eigenvalues).
2. Both $K(\mathbf{p})$ and $H(\mathbf{p})$ are independent of the choice of orthonormal basis.

3.7 Gauss Map in Local Coordinates

In this section we derive formulas for dN , H , and K in terms of the coordinate map $\mathbf{x}$ and the first fundamental form coefficients E, F, G .

Definition 3.19 (Gauss map in local coordinates). We define

$$\mathbf{M}(u, v) := N \circ \mathbf{x}(u, v),$$

viewing the Gauss map in local coordinates. Since $\mathbf{M}_u$ and $\mathbf{M}_v$ lie in $T_p(S)$, which is spanned by $\{\mathbf{x}_u, \mathbf{x}_v\}$, we write

$$\mathbf{M}_u = a_{11}\mathbf{x}_u + a_{21}\mathbf{x}_v, \quad \mathbf{M}_v = a_{12}\mathbf{x}_u + a_{22}\mathbf{x}_v.$$

Lemma 3.20.

$$d(N \circ \mathbf{x}) = d\mathbf{M} = d\mathbf{x} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

which is equivalent to

$$[dN]_{\mathcal{B},\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Since K and H are determined by $[dN]_{\mathcal{B},\mathcal{B}}$, it remains to express a_{ij} in terms of the first and second fundamental form coefficients.

4 Isometries

4.1 Isometries

Definition 4.1 (Diffeomorphism). A diffeomorphism is a map $\phi : S_1 \rightarrow S_2$ such that

- ϕ is smooth (between surfaces),
- ϕ is a bijection (so ϕ^{-1} exists),
- ϕ^{-1} is also smooth (between surfaces).

We say that two surfaces are **diffeomorphic** if there exists a diffeomorphism between them.

Facts.

1. If $\phi : S_1 \rightarrow S_2$ is a diffeomorphism, then its inverse $\phi^{-1} : S_2 \rightarrow S_1$ is also a diffeomorphism.
2. If $\phi : S_1 \rightarrow S_2$ is a diffeomorphism, and $\mathbf{x} : U \rightarrow S_1$ is a coordinate function of S_1 , then

$$\mathbf{y} = \phi \circ \mathbf{x} : U \rightarrow S_2$$

is a coordinate function of S_2 .

Now, we introduce the notion of an **isometry**.

Definition 4.2 (Isometry). A diffeomorphism $\phi : S_1 \rightarrow S_2$ is an **isometry** if the arc lengths of tangent vectors remain unchanged under $d\phi : T_{\mathbf{p}}(S_1) \rightarrow T_{\phi(\mathbf{p})}(S_2)$ for all $\mathbf{p} \in S_1$, i.e.

$$|\mathbf{u}| = |d\phi(\mathbf{u})| \quad \text{for all } \mathbf{u} \in T_{\mathbf{p}}(S_1),$$

or equivalently, for all $\mathbf{p} \in S_1$ and for all $\mathbf{u} \in T_{\mathbf{p}}(S_1)$,

$$I_{\mathbf{p}}(\mathbf{u}) = I_{\phi(\mathbf{p})}(d\phi(\mathbf{u})). \quad (*)$$

Definition 4.3 (Isometry, DoCarmo). A diffeomorphism $\phi : S_1 \rightarrow S_2$ is an **isometry** if for all $\mathbf{p} \in S_1$ and for all $\mathbf{u}, \mathbf{v} \in T_{\mathbf{p}}(S_1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle d\phi(\mathbf{u}), d\phi(\mathbf{v}) \rangle. \quad (**)$$

The surfaces S_1, S_2 are then said to be *isometric*.

Remark 4.4. 1. We have an equivalent definition of isometry if we replace Equation (**) by the seemingly weaker Equation (*). It turns out both are equivalent. We prove this in T9Q2.

2. By definition, an isometry is a diffeomorphism.

4.2 Local Isometries

Definition 4.5 (Local Isometry). A smooth map $\phi : S_1 \rightarrow S_2$ is a *local isometry* at $\mathbf{p} \in S_1$ if there is an open neighbourhood V_1 of $\mathbf{p}$ on S_1 and an open neighbourhood V_2 of $\phi(\mathbf{p})$ on S_2 such that $\phi : V_1 \rightarrow V_2$ is an isometry.

A smooth map $\phi : S_1 \rightarrow S_2$ is a *local isometry* if it is a local isometry at every point on S_1 .

Remark 4.6. Key facts about local isometries.

1. Since $\phi : V_1 \rightarrow V_2$ is an isometry, it is also a diffeomorphism.
2. Every (global) isometry is a local isometry.
3. A local isometry $\phi : S_1 \rightarrow S_2$ is **NOT** necessarily a diffeomorphism. It may not even be a bijection. Hence a local isometry may **NOT** be an isometry.
4. ϕ being a local isometry at a point $\mathbf{p}$ is stronger than saying $I_{\mathbf{p}}(\mathbf{u}) = I_{\phi(\mathbf{p})}(d\phi(\mathbf{u}))$ alone. It means $I_{\mathbf{q}}(\mathbf{u}) = I_{\phi(\mathbf{q})}(d\phi(\mathbf{u}))$ for all points $\mathbf{q}$ in the open neighbourhood V_1 of $\mathbf{p}$ and for all $\mathbf{u} \in T_{\mathbf{q}}(S_1)$.

Proposition 4.7. Let $\mathbf{x} : U \rightarrow S_1$ and $\mathbf{y} : U \rightarrow S_2$ be coordinate functions. Let E_1, F_1, G_1 and E_2, F_2, G_2 denote the coefficients of the first fundamental forms on S_1 and S_2 respectively. Suppose $E_1 = E_2$, $F_1 = F_2$, and $G_1 = G_2$ as functions of $(u, v) \in U$. Then

$$\Phi = \mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U) \rightarrow \mathbf{y}(U)$$

is an isometry.

Proof. Let $\mathbf{p} = \mathbf{x}(u, v)$. Let $\mathbf{w} = a\mathbf{x}_u + b\mathbf{x}_v = d\mathbf{x} \begin{pmatrix} a \\ b \end{pmatrix}$ be a tangent vector at $\mathbf{p}$. Then,

$$d\Phi(\mathbf{w}) = d\Phi d\mathbf{x} \begin{pmatrix} a \\ b \end{pmatrix} = d(\mathbf{y} \circ \mathbf{x}^{-1} \circ \mathbf{x}) \begin{pmatrix} a \\ b \end{pmatrix} = a\mathbf{y}_u + b\mathbf{y}_v.$$

Next we compare

$$I_{\mathbf{p}}(\mathbf{w}) = a^2 E_1 + 2abF_1 + b^2 G_1,$$

$$I_{\Phi(\mathbf{p})}(d\Phi(\mathbf{w})) = I_{\Phi(\mathbf{p})}(a\mathbf{y}_u + b\mathbf{y}_v) = a^2 E_2 + 2abF_2 + b^2 G_2.$$

Since $E_1 = E_2$, $F_1 = F_2$, $G_1 = G_2$, the two first fundamental forms agree. Since $\Phi : \mathbf{x}(U) \rightarrow \mathbf{y}(U)$ is a diffeomorphism, it is an isometry. $\square$

Corollary 4.8. Let $\phi : S_1 \rightarrow S_2$ be a diffeomorphism. Let $\mathbf{x} : U \rightarrow S_1$ be a coordinate function. Then $\mathbf{y} = \phi \circ \mathbf{x} : U \rightarrow S_2$ is a coordinate function on S_2 . Let E_i, F_i, G_i be the coefficients of the first fundamental forms on S_i for $i = 1, 2$. Then the following are equivalent:

1. $E_1 = E_2$, $F_1 = F_2$, $G_1 = G_2$ as functions of $(u, v) \in U$.
2. $\phi = \mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U) \rightarrow \mathbf{y}(U)$ is an isometry.

4.3 Gauss Theorema Egregium

Theorem 4.9 (Theorema Egregium). *Let $\phi : S_1 \rightarrow S_2$ be a local isometry. Let $p_1 \in S_1$ and $p_2 \in S_2$. Let K and $\bar{K}$ be the Gaussian curvatures at p_1 and p_2 respectively. Then $K = \bar{K}$.*

Proof. The main idea of the proof is as follows:

1. Suppose $\phi : S_1 \rightarrow S_2$ is a local isometry.
2. Write the Gaussian curvature K on S_1 in terms of the first fundamental form of S_1 .
3. Similarly $\bar{K}$ on S_2 can be written in terms of the first fundamental form of S_2 .
4. Since ϕ is a local isometry, the first fundamental forms agree locally.
5. Hence $K = \bar{K}$.

Let S be a regular surface and $\mathbf{x} : U \rightarrow S$ a coordinate function. The partial derivatives $\mathbf{x}_{uu}, \mathbf{x}_{uv} = \mathbf{x}_{vu}, \mathbf{x}_{vv}$ can be expressed in the basis $\{\mathbf{x}_u, \mathbf{x}_v, N\}$:

$$\begin{aligned} \mathbf{x}_{uu} &= \Gamma_{11}^1 \mathbf{x}_u + \Gamma_{11}^2 \mathbf{x}_v + L_1 N, & \mathbf{x}_{uv} &= \Gamma_{12}^1 \mathbf{x}_u + \Gamma_{12}^2 \mathbf{x}_v + L_2 N, \\ \mathbf{x}_{vu} &= \Gamma_{21}^1 \mathbf{x}_u + \Gamma_{21}^2 \mathbf{x}_v + \bar{L}_2 N, & \mathbf{x}_{vv} &= \Gamma_{22}^1 \mathbf{x}_u + \Gamma_{22}^2 \mathbf{x}_v + L_3 N, \\ N_u &= a_{11} \mathbf{x}_u + a_{21} \mathbf{x}_v, & N_v &= a_{12} \mathbf{x}_u + a_{22} \mathbf{x}_v. \end{aligned}$$

The coefficients Γ_{ij}^k are the **Christoffel symbols**.

Lemma 4. The Christoffel symbols Γ_{ij}^k are functions of E, F, G and their partial derivatives. In particular,

$$\Gamma_{11}^1 = \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)}, \quad \Gamma_{11}^2 = \frac{-FE_u + 2EF_u - EE_v}{2(EG - F^2)}.$$

Lemma 6.

$$\begin{aligned} EK &= (\Gamma_{11}^2)_v - (\Gamma_{12}^2)_u + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - (\Gamma_{12}^2)^2, \\ FK &= (\Gamma_{12}^1)_u - (\Gamma_{11}^1)_v + \Gamma_{12}^1 \Gamma_{12}^2 - \Gamma_{11}^2 \Gamma_{22}^1. \end{aligned}$$

The proof starts from $(\mathbf{x}_{uu})_v - (\mathbf{x}_{uv})_u = 0$, substitutes the expressions above, applies the product rule and the Weingarten equations, then collects coefficients in $\{\mathbf{x}_u, \mathbf{x}_v, N\}$.

Completing the proof. Let $\mathbf{x} : U \rightarrow S_1$ be a coordinate function; then $\mathbf{y} = \phi \circ \mathbf{x} : U \rightarrow S_2$ is also a coordinate function. Since ϕ is a local isometry, $E = \bar{E}$, $F = \bar{F}$, $G = \bar{G}$ (and all their partial derivatives agree), so $\Gamma_{ij}^k = \bar{\Gamma}_{ij}^k$. By Lemma 6, K depends only on E, F, G and the Christoffel symbols, hence $K = \bar{K}$. $\square$

Remark 4.10. A modification of the proof of Lemma 6 yields the **Mainardi–Codazzi equations**:

$$\begin{aligned} e_v - f_u &= e\Gamma_{12}^1 + f(\Gamma_{12}^2 - \Gamma_{11}^1) - g\Gamma_{11}^2, \\ f_v - g_u &= e\Gamma_{22}^1 + f(\Gamma_{22}^2 - \Gamma_{12}^1) - g\Gamma_{12}^2. \end{aligned}$$

Theorem 4.11 (Fundamental Theorem of Surfaces). *Let E, F, G, e, f, g be smooth functions on an open set U with $EG - F^2 > 0$, $E > 0$, $G > 0$. Form the Christoffel symbols Γ_{ij}^k from E, F, G as in Lemma 4. Suppose E, F, G, e, f, g and Γ_{ij}^k satisfy Lemma 6 and the two Mainardi–Codazzi equations. Then for any $(u_0, v_0) \in U$ there exists a regular surface S and a coordinate function $\mathbf{x} : U' \rightarrow S$ (with $U' \subseteq U$ open, $(u_0, v_0) \in U'$) such that E, F, G and e, f, g are the first and second fundamental forms of S respectively.*

5 Vector Fields & Geodesics

5.1 Tangent Vector Fields

Definition 5.1 (Vector Field). Let S be a regular surface and let $W \subseteq S$ be an open subset. A **vector field** $\mathbf{w}$ on W is a correspondence that assigns to each $p \in W$ a tangent vector $\mathbf{w}(p) \in T_p(S)$.

We can express $\mathbf{w}$ in terms of a coordinate function. More precisely, if $p = \mathbf{x}(u, v) \in S$, then

$$\mathbf{w}(\mathbf{x}(u, v)) = a(u, v) \mathbf{x}_u + b(u, v) \mathbf{x}_v \in T_p(S)$$

for some functions a, b on U . In matrix form,

$$\mathbf{w}(p) = a(u, v) \mathbf{x}_u + b(u, v) \mathbf{x}_v = d\mathbf{x} \begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}.$$

We interpret this expression by viewing $\begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}$ as a vector field on U , and $d\mathbf{x}$ as the map transforming it to the field $\mathbf{w}$ on S .

Definition 5.2 (Smooth Vector Field). We say that a vector field $\mathbf{w}$ is **smooth** at $p_0 = \mathbf{x}(u_0, v_0)$ if the vector field $\begin{pmatrix} a(u, v) \\ b(u, v) \end{pmatrix}$ on U is smooth at (u_0, v_0) .

This definition is independent of the choice of coordinate function.

Proposition 5.3 (Smoothness is independent of coordinate function). *The smoothness of $\mathbf{w}$ at a point p does not depend on the choice of coordinate function used to define it.*

Proof. Suppose $\mathbf{y}: U' \rightarrow S$ is another coordinate function with $p \in \mathbf{y}(U')$. Then

$$\mathbf{w}(q) = a(u, v) \mathbf{x}_u + b(u, v) \mathbf{x}_v = c(\xi, \eta) \mathbf{y}_\xi + d(\xi, \eta) \mathbf{y}_\eta$$

for all q in a neighbourhood of p . Without loss of generality assume $\mathbf{x}(U) = \mathbf{y}(U')$ (otherwise restrict to $U \cap U'$). Suppose $\mathbf{w}$ is smooth with respect to $\mathbf{y}$. One then shows via the chain rule and the transition map $\mathbf{y}^{-1} \circ \mathbf{x}$ that it is also smooth with respect to $\mathbf{x}$. $\square$

5.2 A Good Definition for the Derivative of a Vector Field

Definition 5.4 (Differentiation along a curve). Let $\alpha(t) = \mathbf{x}(u(t), v(t))$ be a curve on S . Write

$$\mathbf{w}(\alpha(t)) = a(u(t), v(t)) \mathbf{x}_u + b(u(t), v(t)) \mathbf{x}_v.$$

If $\alpha(0) = p$ and $\alpha'(0) = (a_1, a_2)$, then by the chain rule,

$$\left. \frac{d}{dt} \mathbf{w}(\alpha(t)) \right|_{t=0} = a_1 \left. \frac{\partial \mathbf{w}}{\partial u} \right|_p + a_2 \left. \frac{\partial \mathbf{w}}{\partial v} \right|_p.$$

Crucially, this expression depends on the curve α *only through its tangent vector* $\alpha'(0) = (a_1, a_2)$.

Differentiating along the line $\alpha(t) = p + t\mathbf{v} = (p_1 + ta_1, p_2 + ta_2)$ recovers the classical directional derivative:

$$\left. \frac{d}{dt} \mathbf{w}(\alpha(t)) \right|_{t=0} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{w}(p_1 + a_1 \Delta t, p_2 + a_2 \Delta t) - \mathbf{w}(p_1, p_2)}{\Delta t}.$$

In general, $\frac{d}{dt} \mathbf{w}(\alpha(t))$ need *not* lie in the tangent plane. This motivates the following definition.

Definition 5.5 (Covariant Derivative). The **covariant derivative** of $\mathbf{w}$ at p relative to the tangent vector $\alpha'(0)$ is

$$\frac{D\mathbf{w}}{dt} = \frac{d}{dt} \mathbf{w}(\alpha(t)) - \left\langle \frac{d}{dt} \mathbf{w}(\alpha(t)), \mathbf{N} \right\rangle \mathbf{N},$$

i.e. the orthogonal projection of $\frac{d}{dt} \mathbf{w}(\alpha(t))$ onto the tangent plane $T_p(S)$.

5.3 Christoffel Symbols in Vector Fields

Christoffel Symbols in the Covariant Derivative

Let $\alpha(t) = (u(t), v(t))$ and write

$$\mathbf{w}(\alpha(t)) = A(t) \mathbf{x}_u + B(t) \mathbf{x}_v, \quad A(t) = a(u(t), v(t)), \quad B(t) = b(u(t), v(t)).$$

Differentiating by the product rule and expanding via the Christoffel symbols, then projecting onto the tangent plane, yields

$$\begin{aligned} \frac{D\mathbf{w}}{dt} &= \left(A' + \Gamma_{11}^1 A u' + \Gamma_{12}^1 A v' + \Gamma_{21}^1 B u' + \Gamma_{22}^1 B v' \right) \mathbf{x}_u \\ &\quad + \left(B' + \Gamma_{11}^2 A u' + \Gamma_{12}^2 A v' + \Gamma_{21}^2 B u' + \Gamma_{22}^2 B v' \right) \mathbf{x}_v. \end{aligned}$$

Remark 5.6. Since the Christoffel symbols Γ_{ij}^k are determined entirely by the first fundamental form coefficients E, F, G (as shown in Chapter 4), the covariant derivative is an *intrinsic* quantity: it can be computed purely from the first fundamental form, without any reference to the unit normal $\mathbf{N}$.

Maps Between Vector Fields

Let S_1, S_2 be regular surfaces, $\mathbf{x}: U \rightarrow S_1$ a coordinate function, and $\phi: S_1 \rightarrow S_2$ a diffeomorphism. Then $\mathbf{y} = \phi \circ \mathbf{x}: U \rightarrow S_2$ is also a coordinate function. For a vector field $\mathbf{w} = a(u, v) \mathbf{x}_u + b(u, v) \mathbf{x}_v$ on S_1 ,

$$d\phi(\mathbf{w}) = a(u, v) \mathbf{y}_u + b(u, v) \mathbf{y}_v,$$

which is a vector field on S_2 .

Proposition 5.7. *Let $\phi: S_1 \rightarrow S_2$ be an isometry. Then*

$$\frac{D(d\phi(\mathbf{w}))}{dt} = d\phi\left(\frac{D\mathbf{w}}{dt}\right).$$

Proof (sketch). Let E_i, F_i, G_i denote the first fundamental form coefficients with respect to $\mathbf{x}$ on S_1 and $\mathbf{y} = \phi \circ \mathbf{x}$ on S_2 . Since ϕ is an isometry, the two first fundamental forms coincide, hence the Christoffel symbols of S_1 and S_2 are identical. The covariant derivative formula above then shows both sides have the same expression in coordinates. $\square$

5.4 Geodesic Curvature

In this section, we introduce the *geodesic curvature*.

Consider a point $\mathbf{p} = \alpha(s_0) \in S$ for some point $s_0 \in U$. Then, we define

$$\mathbf{t} = \alpha'(s) \quad \mathbf{s} = N \wedge \mathbf{t}$$

and these two (unit) vectors form an *orthonormal basis* at $\mathbf{p}$.

And then, we have $\alpha''(s_0)$ to be perpendicular to $\mathbf{t}$, but not necessarily on the tangent plane. So, it is made of components of N and $\mathbf{s}$.

It is easier to visualise with this diagram.

insert diagram here.

Then, we have:

$$\alpha''(s_0) = \langle N, \alpha''(s_0) \rangle N + \langle \mathbf{s}, \alpha''(s_0) \rangle \mathbf{s}$$

And $\langle N, \alpha''(s_0) \rangle$ is the size of the vector that is projected from $\alpha''(s_0)$ onto the "normal plane", which we recall from 3.12 is in fact the *normal curvature*!

Proposition 5.8. *Let k_n be the normal curvature at $\mathbf{v}$. We show that the following are equivalent:*

1. $k_n = \langle N, \alpha''(0) \rangle$
2. $k_n = \Pi_v(v)$

Proof. We know that $\langle N(\alpha(s)), \alpha'(s) \rangle = 0$ since they are perpendicular to each other. So, it follows that

$$\begin{aligned} \langle N(\alpha(s)), \alpha'(s) \rangle = 0 &\iff \frac{d}{ds} \langle N(\alpha(s)), \alpha'(s) \rangle = 0 \\ &\iff \langle N, \alpha''(s) \rangle + \langle dN(\alpha'(s)), \alpha'(s) \rangle = 0 \\ &\iff \langle N, \alpha''(s) \rangle = -\langle dN(\alpha'(s)), \alpha'(s) \rangle = \Pi_{\mathbf{p}}(\alpha'(0)) = k_n \end{aligned}$$

□

And we define the second component as the geodesic curvature.

Definition 5.9 (Geodesic Curvature). We define

$$k_g := \langle \mathbf{s}, \alpha''(s_0) \rangle$$

to be the *geodesic curvature* of $\alpha(s)$ at the point s_0 .

And so, we can simplify the expression for α'' to be

$$\alpha''(s_0) = k_n \mathbf{N} + k_g \mathbf{s}$$

Proposition 5.10. *Let k , k_n and k_g be the curvature, normal curvature and geodesic curvature of α respectively. Then,*

$$k = k_n^2 + k_g^2$$

Proof. Follows from Pythagoras Theorem.

□

5.5 Geodesic

We introduce the concept of a geodesic. First, we motivate the idea using a proposition.

Proposition 5.11. *Let $\alpha(s)$ be a curve on a surface S parameterized by arc length. Then, the following are equivalent.*

For all s ,

1.

$$\frac{D\alpha'}{ds} = (0, 0, 0)$$

2. $\alpha''(s)$ is perpendicular to the tangent space.

3. $\alpha''(s)$ is parallel to the normal vector $\mathbf{N}$.

4. The geodesic curvature k_g of α is zero.

Proof. (1) $\Rightarrow$ (2) :

(2) $\Rightarrow$ (3) :

(3) $\Rightarrow$ (4) :

(4) $\Rightarrow$ (1) :

□

6 Gauss-Bonnet Theorem

– omitted for now –

7 Tutorials

8 Tutorials

9 Cheatsheet
